

CSCI3170

Introduction to Database Systems

4-relational_algebra_derived_v2_20260924

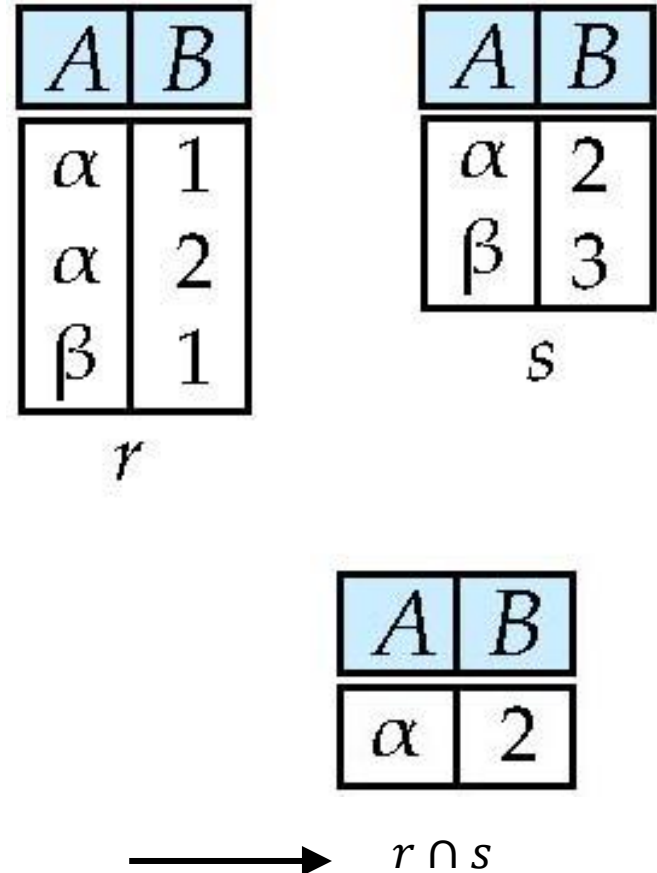
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Part II: Derived Operations

Set intersection, Natural join, Joins, Divide

Set-Intersection

- Notation: $r \cap s$
 - $r \cap s = \{t \mid t \in r \wedge t \in s\}$
- $R \cap S$ returns a relation instance containing all tuples that occur in both R and S
 - R and S must be **union-compatible**
 - schema of result is defined to be identical to the schema of R
- Note: $r \cap s = r - (r - s)$

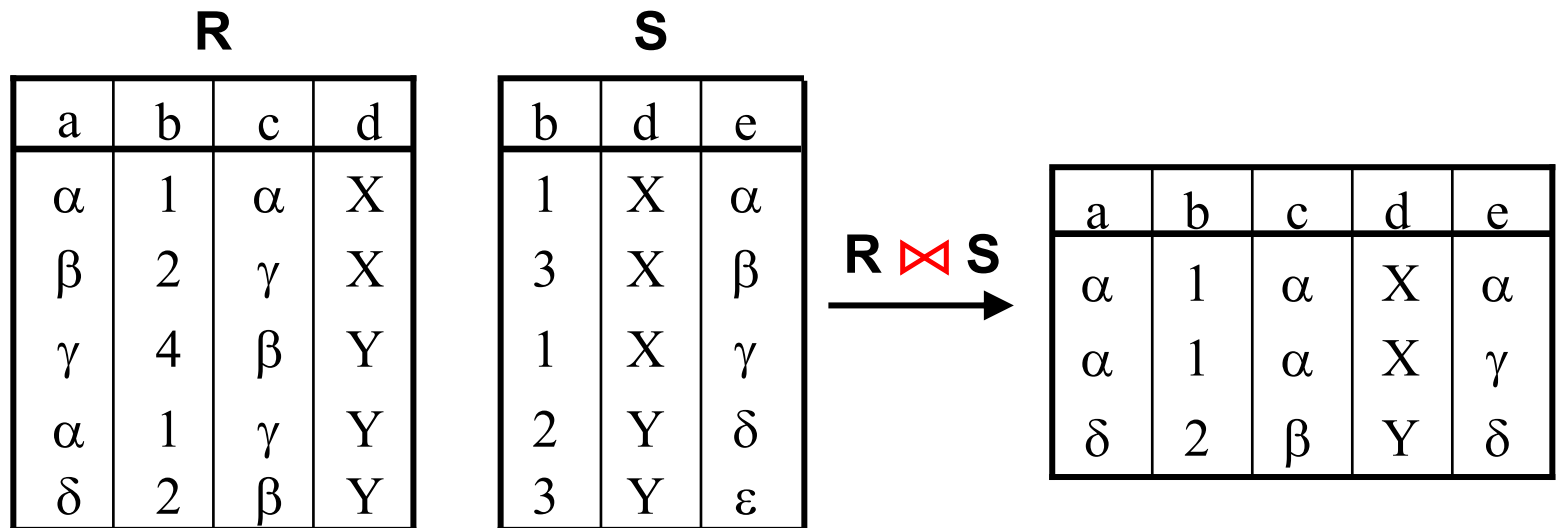


Natural Join

- Notation: $r \bowtie s$
 - Let r and s be relations on schemas R and S respectively.
- $r \bowtie s$ is a relation on schema $R \cup S$ obtained as follows:
 - Consider each pair of tuples t_r from r and t_s from s .
 - If t_r and t_s have the same value on each of the attributes in $R \cap S$, add a tuple t to the result, where
 - t has the same value as t_r on r
 - t has the same value as t_s on s
- Example
 - $R = (A, B, C, D), S = (E, B, D)$
 - Result schema is (A, B, C, D, E)
 - $r \bowtie s$ is defined as:
 - $\Pi_{r.A, r.B, r.C, r.D, s.E}(\sigma_{r.B=s.B \wedge r.D=s.D}(r \times s))$

Natural Join

- Notation for reference: $r \bowtie s$
 - Natural join is **associative**: $(instructor \bowtie teaches) \bowtie course$ equivalent to $instructor \bowtie (teaches \bowtie course)$
 - Natural join is **commutative**: $instructor \bowtie teaches$ equivalent to $teaches \bowtie instructor$



- If the two relations R and S have no attributes in common?
 - then $R \bowtie S$ is simply the cross-product

Theta Join

- $r \bowtie_{\theta} s$ ($r \bowtie_{\theta} s = \sigma_{\theta}(r \times s)$)
 - Combines selection and Cartesian product into a single operation
 - A general version of join operation(s): accepts any arbitrary comparison for *join condition* θ
 - **equijoin** is theta join when using equality

S

sid	sname	rating	age
22	Dustin	7	45.0
31	Lubber	8	55.5
58	Rusty	10	35.5

R

sid	bid	day
22	101	10/10/96
58	103	11/12/96

$S \bowtie_{S.sid=R.sid} R$

sid	sname	rating	age	sid	bid	day
22	Dustin	7	45.0	22	101	10/10/96
58	Rusty	10	35.5	58	103	11/12/96

Because pure relational algebra doesn't natively support "dot notation" (d.salary), doing **any Cartesian product/ join** like this technically creates a schema collision error in a strict theoretical environment because the columns aren't unique. (see formal practice next page)

Problem! Formal relational algebra does not allow duplicate column names in a relation.

When joining, you must explicitly **rename** the colliding columns before hand. **See next slide.**

If we are strict with pure relational algebra, and put aside dot-nations, there is a real problem.

Theta Join (and rename to resolve clash)

- $r \bowtie_{\theta} s$ ($r \bowtie_{\theta} s = \sigma_{\theta}(r \times s)$)
 - Combines selection and Cartesian product into a single operation
 - A general version of join operation(s): accepts any arbitrary comparison for *join condition* θ
 - **equijoin** is theta join when using equality

S

sid	sname	rating	age
22	Dustin	7	45.0
31	Lubber	8	55.5
58	Rusty	10	35.5

R

sid	bid	day
22	101	10/10/96
58	103	11/12/96

$$(\rho_{s_id \leftarrow sid}(S)) \bowtie_{s_id=r_id} (\rho_{r_id \leftarrow sid}(R))$$

s_id	sname	rating	age	r_id	bid	day
22	Dustin	7	45.0	22	101	10/10/96
58	Rusty	10	35.5	58	103	11/12/96

When joining,
we explicitly
renamed the
colliding columns
before hand.

Better!

Also a Theta Join

- allows for arbitrary comparison (such as $\geq$)

In pure, theoretical relational algebra, relations (tables), the rename operator is traditionally used to change column names and avoid ambiguity.

S

sid	sname	rating	age
22	Dustin	7	45.0
31	Lubber	8	55.5
58	Rusty	10	35.5

R

sid	bid	day
22	101	10/10/96
58	103	11/12/96

$$(\rho_{s_id \leftarrow sid}(S)) \bowtie_{s_id < r_id} (\rho_{r_id \leftarrow sid}(R))$$

s_id	sname	rating	age	r_id	bid	day
22	Dustin	7	45.0	58	103	11/12/96
31	Lubber	8	55.5	58	103	11/12/96

Formal relational algebra does not allow duplicate column (attribute) names within a single relation. When joining, you must explicitly **rename** the colliding columns before hand

Theta Join, Equijoin and Natural Join?

- In this case, the theta join might behave exactly like a natural join.
Not all Equi-joins are Natural joins.

R

a	b	c
α	1	α
β	5	γ
γ	4	β
α	1	γ
δ	2	β

S

b	e	f
1	X	α
3	X	β
1	X	γ
2	Y	δ
3	Y	ϵ

$\Pi_{a,R.b,c,e,f} (R \bowtie_{R.b=S.b} S)$

a	b	c	e	f
α	1	α	X	α
α	1	α	X	γ
α	1	γ	X	α
α	1	γ	X	γ
δ	2	β	Y	δ

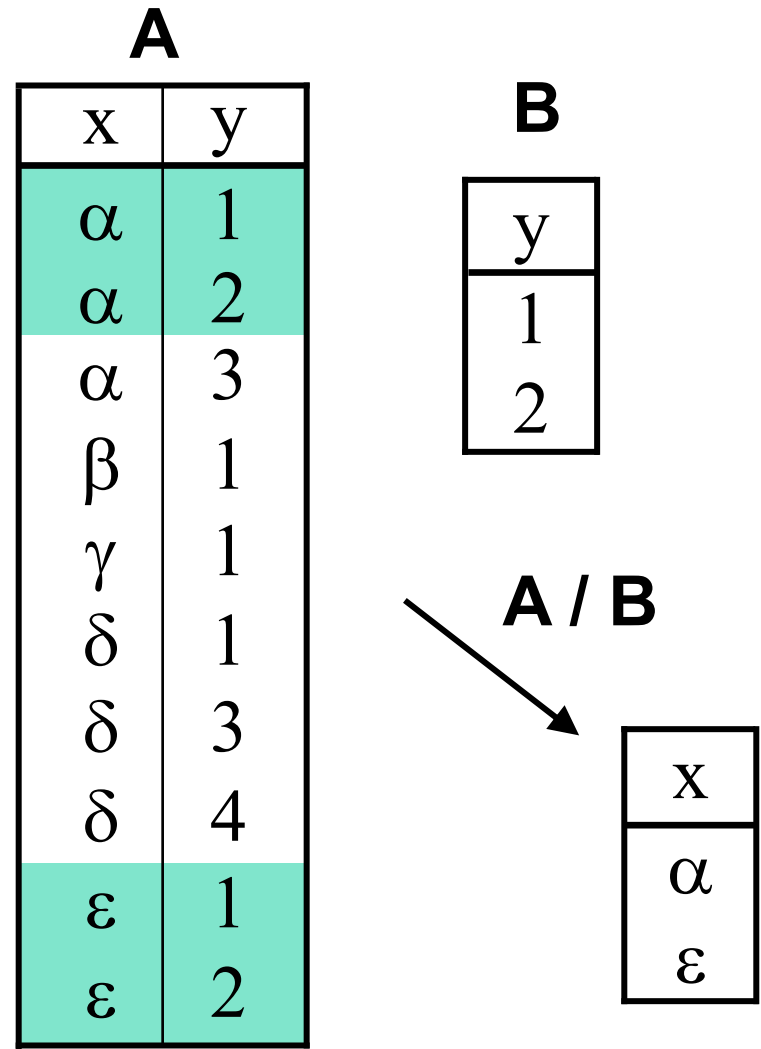
For simplicity we will have to refer to the dot notation sometimes.

But note that pure relational algebra doesn't natively support "dot notation" (d.salary), if there are clashes, assume the rename operator is traditionally used to change column names and avoid ambiguity; rather than dot notations commonly seen in practice.

(try to express the “largest salary” expression again without dot notation?)

Division (/)

- Simple Example
 - Consider two relations A and B
 - A has exactly two fields x and y
 - B has just one field y, with the same domain as in A.
 - A/B is set of all x values such that for every y value in a tuple of B, there is a tuple $\langle x, y \rangle$ in A
- X need not be a unary tuples.
- Typically useful for expressing certain kinds of queries, for example, “find the names of sailors who have reserved all boats”.



Division Example

- Query: Retrieve names of employees who work on all the projects every 'John Sung' works on.

Employee

fname	lname	id	bdate	address	salary	sid	dno
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Works_on

id	pno
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- $Sung \leftarrow \sigma_{fname="John" \wedge lname="Sung"} (Employee)$
- $Sung_pnos \leftarrow \Pi_{pno} (Works_on \bowtie Sung)$

fname	lname	id	bdate	address	salary	sid	dno	pno
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- $Result_id \leftarrow Works_on / Sung_pnos$
- $Result \leftarrow \Pi_{fname, lname} (Result_id \bowtie Employee)$

Reference

- Some slides adapted from Database System Concepts – 6th Edition by Silberschatz, Korth and Sudarshan
- Remark: Pure mathematical relational algebra does not formally include dot-notation.

Practice: Sample Schema

- Consider relational schema below, the primary key fields are underlined (**also bolded for your convenience**)

Sailors (sid, sname, age)
Boats (bid, bname, color)
Reserves (sid, bid, date)

Sailors

Sid	Sname	Rating	Age
22	Dustin	7	45
29	Brutus	1	33
31	Lubber	8	55.5
32	Andy	8	25.5
58	Rusty	10	35
64	Horatio	7	35
71	Zorba	10	16
74	Horatio	9	40
85	Art	3	25.5
95	Bob	3	63.5

Boats

bid	bname	color
101	Interlake	blue
102	Interlake	red
103	Clipper	green
104	Marine	red

Reserves

sid	bid	day
22	101	1998-10-10
22	102	1998-10-10
22	103	1998-10-8
22	104	1998-10-7
31	102	1998-11-10
31	103	1998-11-6
31	104	1998-11-12
64	101	1998-9-5
64	102	1998-9-8
74	103	1998-9-8

Figure 1: Instances of Sailors, Boats and Reserves

Practice: Example (1)

- Query 1: Find names of sailors who have reserved boat with bid = 103
- Solution 1.1:
 - $\Pi_{\text{sname}} ((\sigma_{\text{bid}=103} \text{Reserves}) \bowtie \text{Sailors})$
- Solution 1.2:
 - $\rho (\text{Temp1}, \sigma_{\text{bid}=103} \text{Reserves})$
 - $\rho (\text{Temp2}, \text{Temp1} \bowtie \text{Sailors})$
 - $\Pi_{\text{sname}} (\text{Temp2})$
- Solution 1.3:
 - $\Pi_{\text{sname}} (\sigma_{\text{bid}=103} (\text{Reserves} \bowtie \text{Sailors}))$

Sailors (sid, sname, age)
Boats (bid, bname, color)
Reserves (sid, bid, date)

Practice: Example (2)

- Query 2: Find names of sailors who have reserved at least a red boat
- Solution 2.1:
 - $\Pi_{\text{sname}} ((\sigma_{\text{color}='red'} \text{Boats}) \bowtie \text{Reserves} \bowtie \text{Sailors})$
- Solution 2.2:
 - $\Pi_{\text{sname}} (\Pi_{\text{sid}} ((\Pi_{\text{bid}} \sigma_{\text{color}='red'} \text{Boats}) \bowtie \text{Reserves}) \bowtie \text{Sailors})$

Sailors (sid, sname, age)
Boats (bid, bname, color)
Reserves (sid, bid, date)

Practice: Example (3)

- Query 3: Find names of sailors who have reserved at least a red or a green boats.
- Solution 3.1:
 - $\rho (\text{Tempboats}, (\sigma_{\text{color}='red' \vee \text{color}='green'} \text{Boats}))$
 - $\Pi_{\text{sname}} (\text{Tempboats} \bowtie \text{Reserves} \bowtie \text{Sailors})$
- Solution 3.2:
 - $\Pi_{\text{sname}} ((\sigma_{\text{color}='red'} \text{Boats}) \bowtie \text{Reserves}) \bowtie \text{Sailors}) \cup$
 - $\Pi_{\text{sname}} ((\sigma_{\text{color}='green'} \text{Boats}) \bowtie \text{Reserves}) \bowtie \text{Sailors})$

Sailors (sid, sname, age)
Boats (bid, bname, color)
Reserves (sid, bid, date)

Practice: Example (4)

- Query 4: Find names of sailors who have reserved at least a red boat and at least a green boat
- Solution 4:
 - $\rho (\text{Tempred}, \Pi_{\text{sid}} ((\sigma_{\text{color}='red'} \text{Boats}) \bowtie \text{Reserves}))$
 - $\rho (\text{Tempgreen}, \Pi_{\text{sid}} ((\sigma_{\text{color}='green'} \text{Boats}) \bowtie \text{Reserves}))$
 - $\Pi_{\text{sname}} ((\text{Tempred} \cap \text{Tempgreen}) \bowtie \text{Sailors})$
- Note: sid is a key for Sailors, but sname is not a key

Sailors (sid, sname, age)
Boats (bid, bname, color)
Reserves (sid, bid, date)

Practice: Example (5)

- Query 5: Find names of sailors who have reserved every boats
- Solution 5:
 - $\rho (\text{Tempsids}, (\Pi_{\text{sid,bid}} \text{Reserves}) / (\Pi_{\text{bid}} \text{Boats}))$
 - $\Pi_{\text{sname}} (\text{Tempsids} \bowtie \text{Sailors})$
- What if we simply do $\text{Reserves} / (\Pi_{\text{bid}} \text{Boats})$?

Sailors (sid, sname, age)
Boats (bid, bname, color)
Reserves (sid, bid, date)

Practice: Example (6)

- Query 6: To find sailors who have reserved all red boats:
- Solution 6:
 - $\rho (\text{ Tempsids}, (\Pi \text{ sid}, \text{bid Reserves}) / (\Pi \text{ bid } (\sigma \text{ color} = \text{'red'} \text{ Boats}))$
 - $\Pi \text{ sname } (\text{ Tempsids } \bowtie \text{ Sailors})$

Sailors (sid, sname, age)
Boats (bid, bname, color)
Reserves (sid, bid, date)